

Functions and Data Structures

- Video: Functions 5 min
- Video: Variable scope 4 min
- Reading: Function and variable scope 13 min
- Reading: What are data structures? 10 min
- Video: Lists 3 min
- Video: Tuples 3 min
- Video: Sets 4 min
- Video: Dictionaries 6 min
- Video: Inmaps 2 min
- Practice Quiz: Functions, loops and data structures 5 questions
- Reading: Choosing and using data structures 15 min
- Reading: Visual Studio Code on Coursera 10 min
- Programming Assignment: Functions, loops and data structures 3h
- Practice Quiz: Knowledge check: Functions and Data Structures 5 questions
- Reading: Additional resources 5 min

Errors, exceptions and file handling

Visual Studio Code on Coursera

In addition to having Visual Studio Code installed on your own computer, in this course and throughout this program, you'll have the opportunity to work in Visual Studio Code right here on Coursera!

As you progress through the course, you'll be able to write code in hands-on activities called **Labs**. In these labs you'll be able to open Visual Studio Code and start writing code without ever leaving the course.

You'll have plenty of opportunities to see Labs in action later in the course, but for now, use the images below as a visual guide to how Labs will look and operate in your browser.

Lab: Creating an HTML Document

The Labs contain instructions explaining the coding task.

Creating an HTML Document

Open Lab

Instructions

Introduction

In this ungraded lab you will practice creating a simple HTML document.

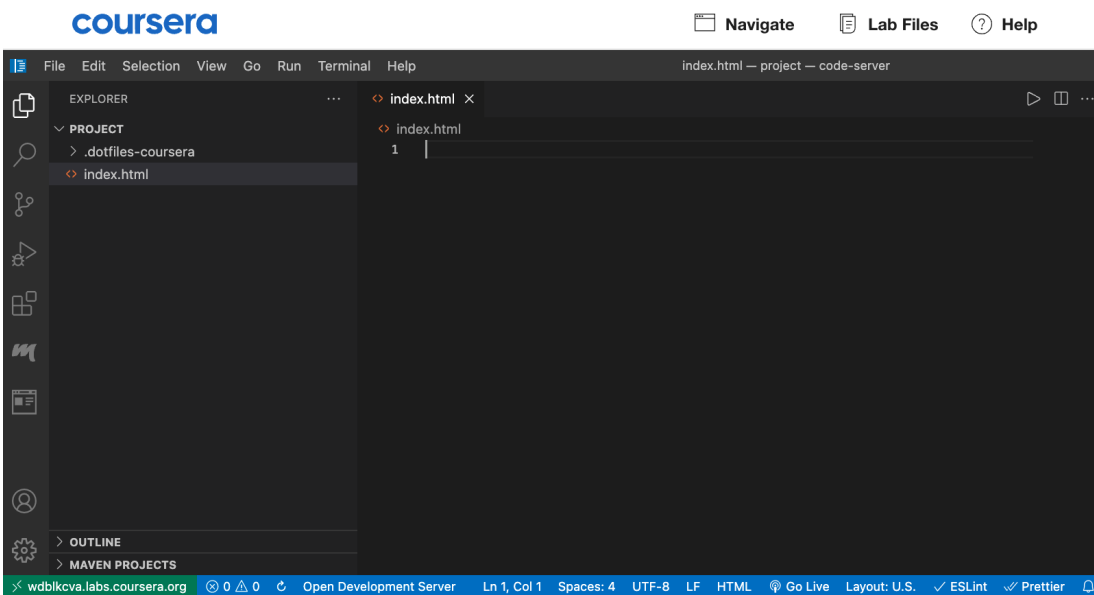
Goal

- Create a valid HTML document that displays a piece of text.

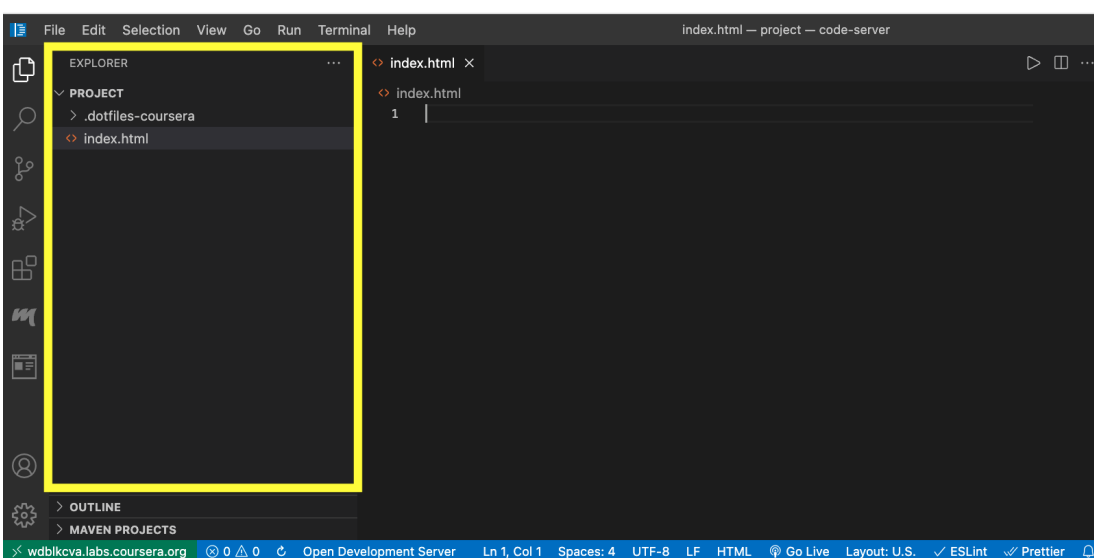
Objectives

- Add the DOCTYPE.
- Add the HTML, head and body elements.
- Add the title element.
- Add the text to the body element.

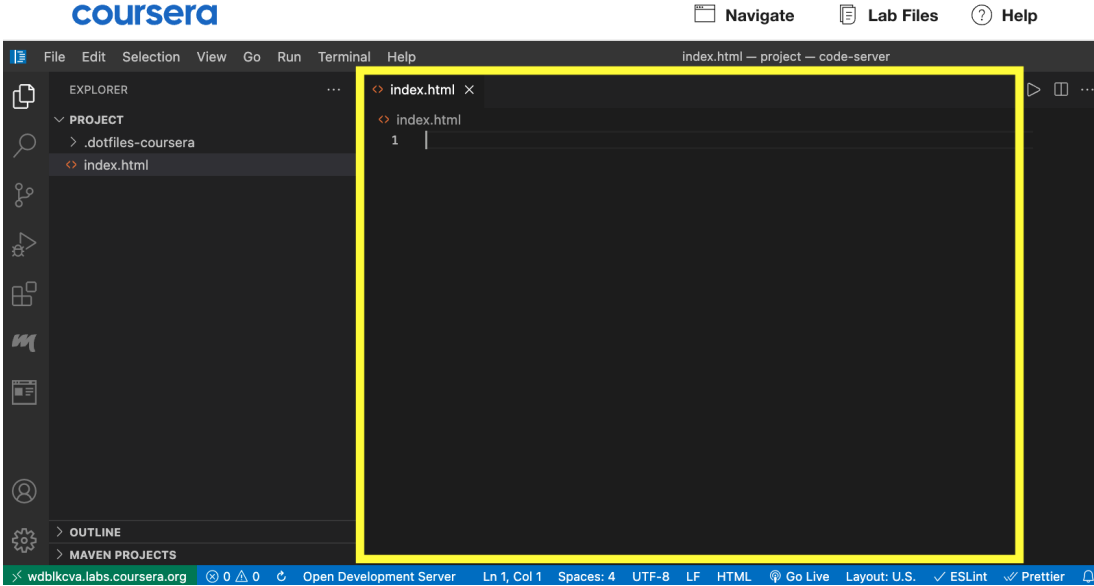
When you click the button to open the lab, a new tab will open with Visual Studio Code already setup and ready for you to start writing code!



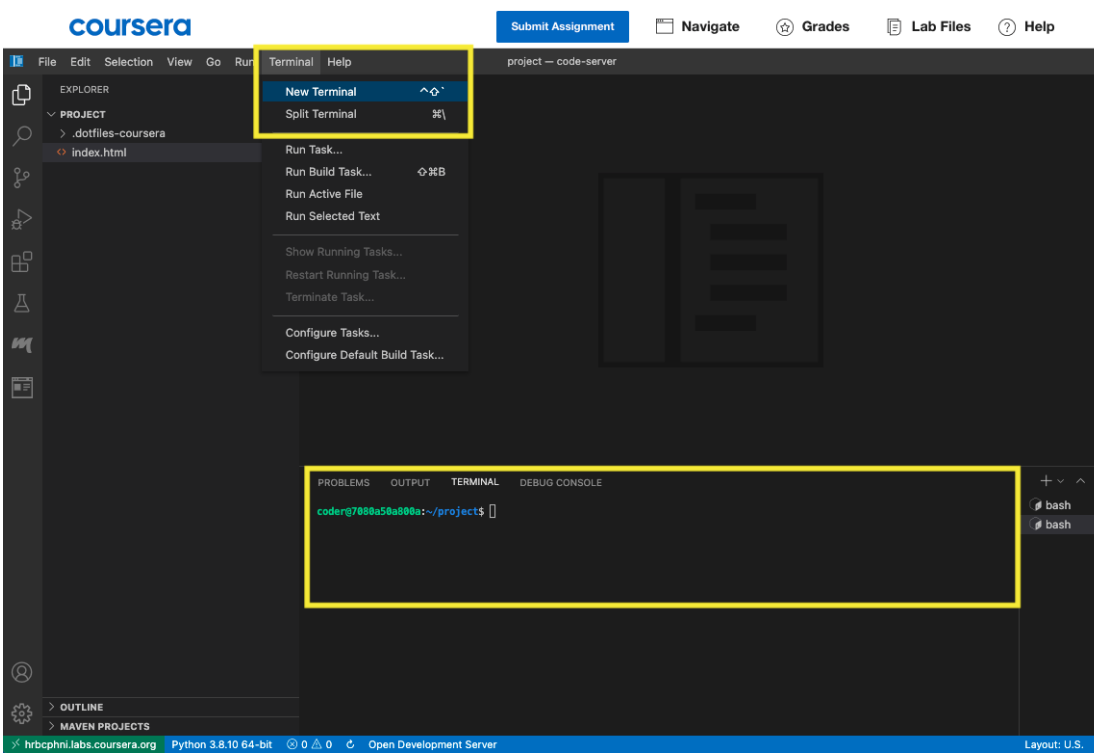
You'll see all the files for the lab in the Project folder in the left sidebar.



And the large editor area where you write your code for the lab.

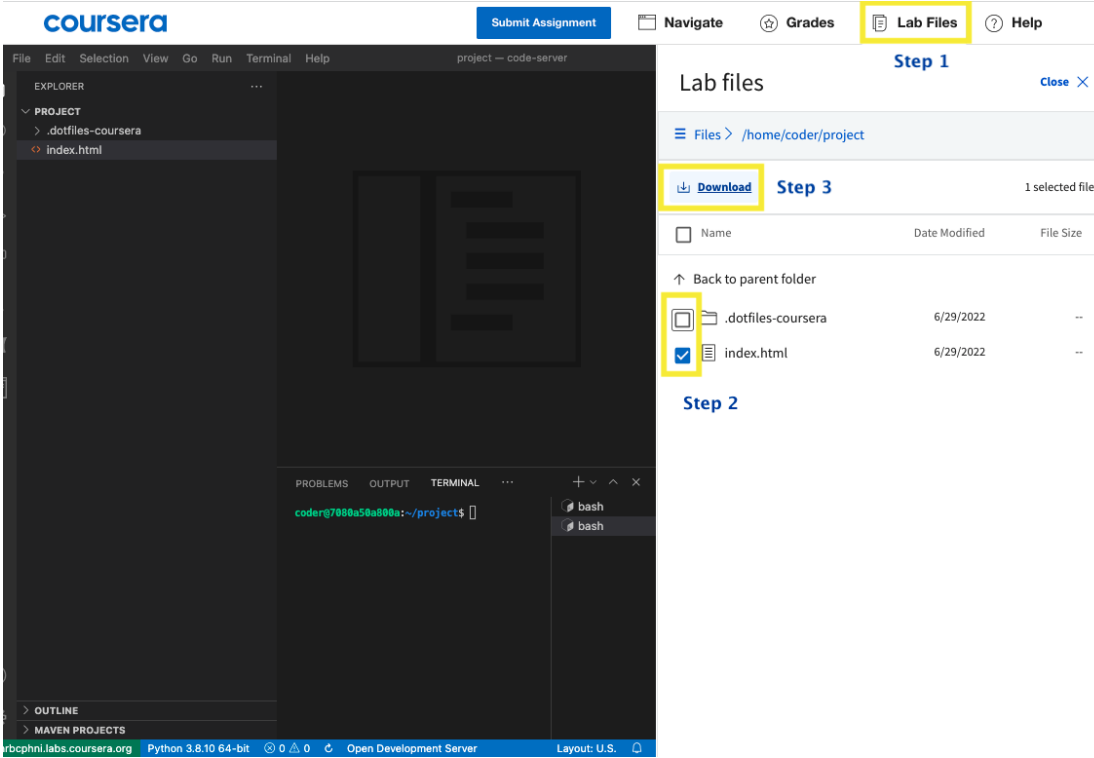


You may need to use a tool called the Terminal from time to time to complete course activities. You can open this by selecting the **Terminal** option in the upper Visual Studio Code toolbar.



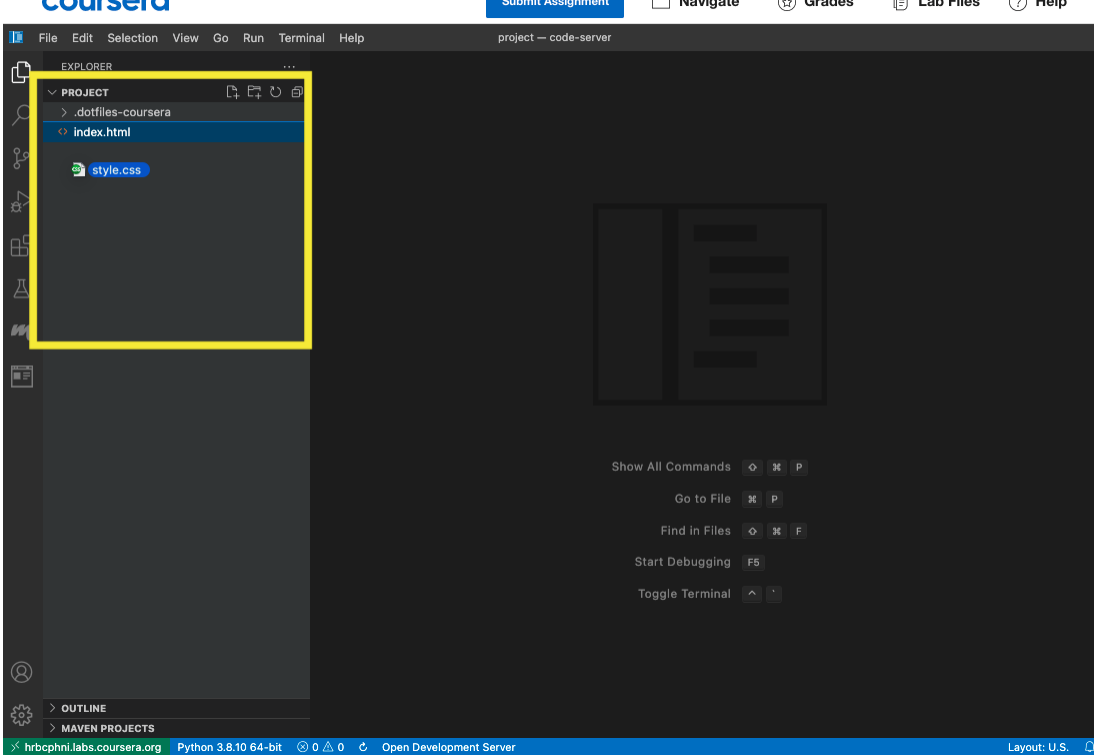
How to download files from your Visual Studio Code Lab to your local device

1. Select the **Lab Files** button in your Lab Toolbar.
2. You'll be able to download your full workspace, specific folders, or individual files through the checkbox selection tool.
3. After you've selected these files, use the **Download** link to download your files to your local device.



How to upload local files to your Visual Studio Code Lab

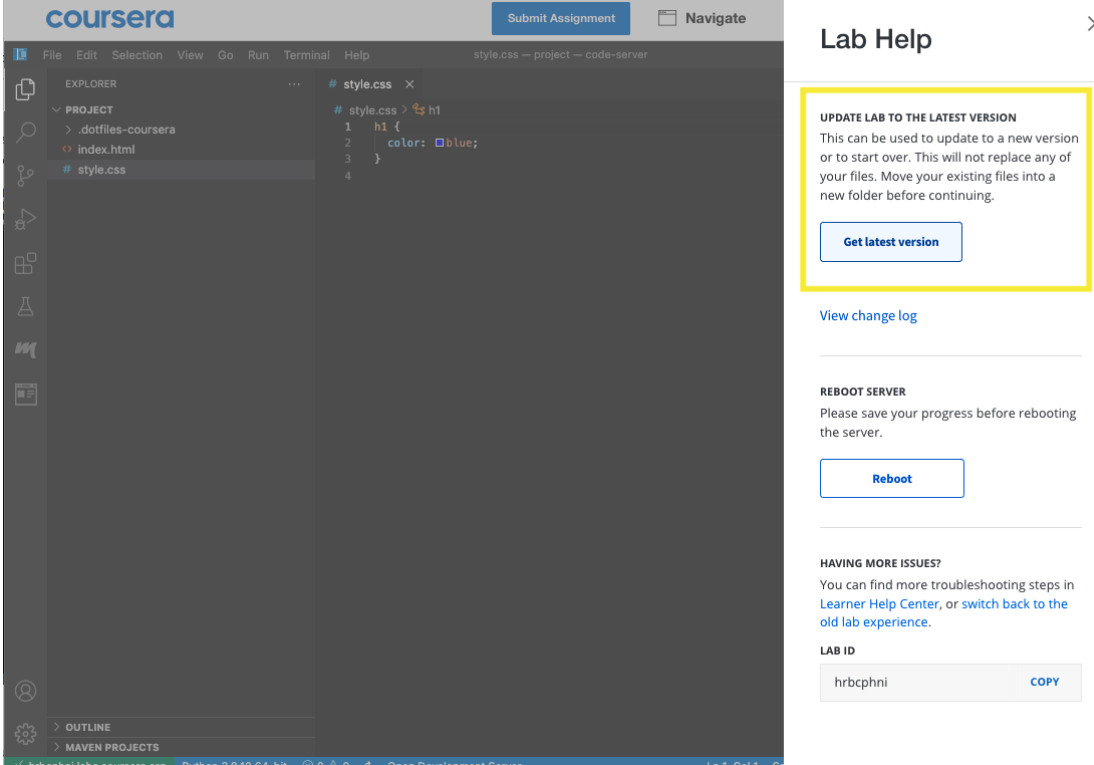
If you'd like to upload your course files from your local device to your Visual Studio Code lab, **drag and drop** your file from your local device into the Visual Studio Code file tree.



How to get a fresh copy of course-provided starter files

Your work will be saved and persist within your Visual Studio Code lab while you are enrolled in the course. If you'd like to get a fresh copy of the original instructor-provided files at any time, you can do this through the **Lab Help** option in your Lab Toolbar. Don't worry - your original work and files will still remain in your lab until you personally remove or delete them, even when refreshing your files through the steps below.

1. First rename your original files to something like `{yourfilename}_{original}{your file extension}`. You can do this by right-clicking on your file in the Visual Studio Code file tree, selecting **Rename**, and providing a new file name.
 - For example for `index.html`, this could be renamed to `'index_{original}.html'`
2. Select **Lab Help** from your Lab Toolbar and then select **Get latest version**.



3. You should now see a fresh copy of the original instructor-provided files in your lab, in addition to your own (renamed) files.

Completed Go to next item

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